UPAR Challenge 2027 — Track 1: Pedestrian Attribute Recognition

Student(s): Haris Ahmed, Syed Huzaifa Ali
Course: Computer Vision
Instructor: Sir Bilal Ahsan
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1. Task and Approach

For this assignment, we selected **UPAR Challenge 2027 Track 1: Pedestrian Attribute Recognition**. The task is to predict **40 binary pedestrian attributes** from surveillance-style pedestrian images. The attributes describe properties such as age group, gender, hair length, upper- and lower-body clothing, clothing colors, and accessories including backpacks, bags, glasses, and hats.

The main difficulty of the challenge is **domain shift**. The hidden Codabench test images come from a different domain than the development data, so the model must generalize to changes in camera quality, viewpoint, illumination, resolution, and background.

Our solution uses an **ImageNet-pretrained ResNet-50**. The original final classification layer was replaced with a 40-output layer, and the network was fine-tuned on the official UPAR Track 1 training split.

2. Methodology

The provided dataset contains **97,669 training images** and **33,407 validation images**. Each image has a 40-dimensional binary label vector.

The dataset is highly imbalanced. For example, `Age-Adult` appears in about **93.47%** of the training set, while `LowerBody-Color-Purple` appears in only **0.17%**. Some attributes also shift noticeably between training and validation, for example `Age-Young` changes from **4.66%** to **1.10%**, and `LowerBody-Length-Short` changes from **20.54%** to **37.51%**.

The model pipeline was:

```
Pedestrian Image → Resize 256×128 → ResNet-50 → 40 logits → Sigmoid → 40 probabilities
```

Training used random horizontal flipping, mild color jitter, ImageNet normalization, and mixed-precision GPU training. The main training settings were:

- **Loss:** BCEWithLogitsLoss
- **Optimizer:** AdamW
- **Learning rate:** 1e-4
- **Weight decay:** 1e-4
- **Batch size:** 32
- **Epochs:** 10
- **GPU:** NVIDIA GeForce RTX 4050 Laptop GPU, 6 GB VRAM
- **Model parameters:** 23,589,992

The best checkpoint was selected using the official `Challenge_Avg` validation metric.

3. Evaluation and Results

The challenge reports `mA`, `Label_F1`, `Instance_Acc`, `Instance_Prec`, `Instance_Rec`, `Instance_F1`, and `Challenge_Avg`. The main ranking metric is `Challenge_Avg`, which is the harmonic mean of `mA` and `Instance_F1`. An attribute is counted as present when its predicted probability is above 0.5.

The best validation checkpoint was obtained at **epoch 8**.

Metric	Best Validation Score
mA	0.807321
Label_F1	0.692363
Instance_Acc	0.774283
Instance_Prec	0.874596
Instance_Rec	0.853393
Instance_F1	0.863864
Challenge_Avg	**0.834636**

The validation `Challenge_Avg` improved from **0.8063** at epoch 1 to **0.8346** at epoch 8. After epoch 8, the score did not improve further. Training loss continued to decrease while validation loss increased, which indicates the beginning of overfitting.

The final model was submitted to the official Codabench Track 1 hidden test set.

- **Submission ID:** 1004995
- **Status:** Finished
- **Official hidden-test score: 0.7021**

The drop from **0.8346 validation** to **0.7021 hidden-test** suggests that the model was affected by the domain shift between the provided development data and the unseen surveillance domain.

4. Challenges Encountered

The main challenges were:

- **Class imbalance:** rare clothing colors and accessories are difficult to learn reliably.
- **Domain shift:** the hidden test domain differs from the development datasets.
- **Small visual details:** glasses, hats, hair length, and rare colors can be difficult to recognize in low-resolution images.
- **Overfitting:** validation loss increased in later epochs even though training loss kept decreasing.
- **Dataset setup:** the official downloader required small Windows compatibility fixes, including a `gdown` version issue and resume-safe handling of downloaded datasets.
- **Codabench packaging:** the final inference package had to run offline with the model weights included in the ZIP and only the allowed runtime dependencies.

5. Possible Improvements

Future work could focus on **class-weighted BCE**, focal loss, or asymmetric loss to improve rare-attribute learning. Stronger domain-generalization augmentation such as random erasing could also be tested. More modern backbones such as ConvNeXt or Vision Transformers, attribute-specific attention, ensembling, and probability calibration may further improve hidden-test performance.

Conclusion

A complete UPAR Track 1 pedestrian attribute recognition pipeline was implemented using an ImageNet-pretrained ResNet-50. The best local validation `Challenge_Avg` was **0.8346**, while the official Codabench hidden-test score was **0.7021**. The difference between the two highlights the difficulty of domain generalization in real-world surveillance data.